

Sudesh Acharya



- » **Fields:** Software Development, Data Science, and Deep Learning.
- » **Stacks:** Python3, HTML5 | CSS3 | JS, MongoDB | MonetDB, C/C++, Basic Rust.
- » **Utils:** RTOS, DjangoREST, Numpy, Pandas, Matplotlib, PyTorch, OpenCV, AWS | GCP, Gymnasium, Linux, Git, Docker, ROS2, etc.
- » **Enjoys:** Knowledge-exchange, Music, Learning new stuffs, Seasonal Hobbies.
- » **Languages:** German: Fluent, Hindi: Fluent, Nepali: Native, Russian: Beginner

»»» Summary

Spirited Developer ready, to offer acquired Software(Instruction, Test-driven) and Machine Learning(Data-driven) development skills and experiences, to spend effort on exciting and fulfilling projects, seeking to undertake such roles and work on innovative and SOTA solutions.

»»» Experiences Timeline

2025.03-now	Embedded C Developer - Kathmandu, Nepal	imark Digital
	» Development of Payment application for Embedded RTOS based PoS systems » Implementation of DQR/Card based payment, HTTP, MQTT, and Other communication Stacks for reliable communication.	
2023.11-24.08	Data Science Researcher Intern - Munich, Germany	FfE e.V.
	» Development of a Prosumer Reinforcement Learning Agent for Energy Cost Optimization. » Using Gymnasium and StableBaselines3 , based on PPO, SAC TD3 Algorithms.	
2022.07-23.08	API server Development with DjangoREST - Karlsruhe, Germany	CloudFluid GmbH
	» Development of Cloud(GCP, AWS) Interface[API] for fluid Cloud-based Simulation Service. » DRF based Django User Auth., Service Backend, Logging(structlog), and Testing(PyTest)	
2021.10-22.03	Data Analytics and Reporting - Rostock, Germany	VestiFi GmbH
	» Generation of Visualization(Matplotlib, Seaborn) and Reporting(LaTeX, Jinja). » Extraction(MonetDB SQL), Transformation(Numpy), and Loading(MongoDB) pipeline for wifi radio pcaps.	
2021.05-22.02	Lidar Data Analysis, and ETL - Kiel, Germany	FuE FH Kiel GmbH
	» Raw maritime 3D Lidar data collection and management within Kiel Förde » Analysis and Transformation of raw Lidar data to several point cloud formats(.pcd, .npy)	
2021.(05-06)	NLP/NLU Development Internship - Munich, Germany	ROKIN GmbH
	» Document Classification(tagging) with BERT and Derivatives » NLP based Model Development, Evaluation, Article prediction and Testing using Google Cloud Platform (GCP)	
2017.(02-05)	ODOO Module Programmer - Lalitpur, Nepal	BI Solutions Pvt. Ltd.
	» ODOO ERP and CRM Custom Module Development » Python /ORM /MVC, PostgreSQL, Scrum/Kanban Project Mgmt.	

»»» Academic Timeline

2012 - 2016	Bachelor's Degree	RGPV University, India
	» Computer Science and Engineering » Theory of Computation, Data Structures and Algorithms, Digital Circuits and System etc.	

2010 - 2012	High School	Caribbean HSS, Nepal
	» Major : Natural and Formal Science » Modern Physics, Biology, Chemistry, and Mathematics	
»» Personal and Academic Projects		
2020 - 2021	Exploration of Art Generation using Deep Generative Models	PyTorch
	» Exploration and Evaluation of Generative Models in Art Generation Domain. » Progressive Implementaiton and Observation on (Unconditional and Conditional) GANs	
2020 - 2021	Classification of Thoractic diseases using Deep Learning	Keras Tensorflow
	» Implementation, Optimization and Evaluation on Custom CNNs.) » Pre-training and Fine Tuning(Transfer Learning) using Densenet121 Architecture	
2019- 2020	Time Series Analysis and Forecasting for Energy Prosumption	Python, R
	» Time Series Analysis and Forecasting using Classical ARIMAs and ML algorithms » On an hourly energy production and consumption data from <i>energycharts.info</i>	
2019- 2020	Model Selection,Data Exploration and Visualization	Scikit-Learn
	» High-dimensional Data Exploration: PCA, LDA, t-SNE, ISOMAP etc. » Model Selection, Validation and Evaluation using criterias (AIC, BIC, MDL)	
2018 - 2019	Classical Machine Learning Implementation	Scikit-Learn
	» Clustering, Classification, Regression: Support Vectors, Perceptron, DecisionTree, KNN) » Data: UCI banknote authentication, Segmentation, Mice Protein data Clustering, etc.	
»» Certification Courses		
2020 - 2021	Python for Data Science, AI and Development	IBM Coursera
	» Fundamentals of Python Programming for Analytics » Data Analysis with Numpy,Pandas, Matplotlib, and Seaborn	
2020 - 2021	Deep Learning Specialization	Coursera
	» Deep Learning Fundamentals, Computer Vision, Sequence and Attention Models(NLP, NLU) » Optimization, Hyperparameter Tune/Search, and Regularization.	
2020	AI for Medical Diagnosis	Deeplearning.ai
	» Disease detection and classification using Convolutional Neural Networks » Evaluation metrics, domain challenges with medical datasets	

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Github : github.com/mnpr-vcs/

